Mathieu**Dugré**

Software Developer

Contact

6590A rue Louis-Dupire
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mathdugre

Languages

French native
English bilingual

Programming

Python, Dask, Bash
Spark, Docker, Linux

Soft skills

leadership,
management,
communication,
problem solving

Education

09/17–05/20 **B.Comp.Sci.** Honours Computer Aplications — Concordia University, Montreal, QC
Major in Mathematics and Statistics.
· Honours work supervised by Dr. Tristan Glatard on a project aiming at exploring the possible strategies to minimize the data transfer time of containerized application for High Performance Computing.

Experience

Work experience

05/19–now **Concordia University** — Montreal, QC
Research Assistant
· Conducted benchmark experiments and wrote a paper on the performance of two Big Data engines: Dask and Apache Spark.
· Worked on the development of tools and procedures to facilitate the sharing of restricted data from PERFORM Center onto the Canadian Open Neuroscience Platform (CONP).
· Exploring different strategies to pinpoint bottleneck in application executed on High Performance Computing environment.

Teaching Experience

2019 **MTL - Brainhack Global 2019** — Montreal, QC
Workshop demonstrator
Taught a workshop on the Git and GitHub workflow in the scope of research in neuroimaging.

Extracurriculars

09/19–now **Seminary on Undergraduate Mathematics in Montreal** — Montreal, QC
VP Communication
· Developed the website for the event.
· Designed the process for inscription and submission for talk and poster.
· Wrote funding requests.

05/19–05/20 **Concordia Software Engineering And Computer Science Society** — Montreal, QC
Co-VP Competitions
· Formed teams for inter-university competitions
· Organized and created competition tryouts

08/18–05/19 **Data Innovation Playground** — Montreal, QC
Director of projects
· Mentored students on building data science projects on a weekly basis.
· Recruited mentors, speakers, and sponsors for The Innovative Society.
· Built the website for our rebranding.

Competitions

11/19	**International Collegiate Programming Contest (ICPC) - NENA Regional** 10th place	Rochester, NY
09/18	**IEEExtreme 12.0** 10th place (Canada)	Montreal, QC
04/18	**Google Tech Challenge** 1st place	Montreal, QC
09/17	**IEEExtreme 11.0** 15th place (Canada)	Montreal, QC

Awards

05/19	**Supplements of the NSERC Undergraduate Student Research Awards**	Concordia University
08/19	**NSERC Undergraduate Student Research Awards (USRA)**	Concordia Univeristy

Interests

Professional: Big Data, Distributed System, Parallel Computing, Data Science, DevOps, High Performance Computing, Reproducibility, Algorithms Design.

Personal: Cooking, Reading, Competitive Programming, Martial Arts.

Publications

Articles in peer-reviewed journals

1. A performance comparison of Dask and Apache Spark for data-intensive neuroimaging pipelines
   Workflows in Support of Large-Scale Science (Nov. 2019). Super Computing, Presentation.

Published code

For an up-to-date list of published code projects, please visit my GitHub profile at https://github.com/mathdugre.